Presented by :-Akshata Gaikwad (2021MAS1005)
Manisha Sehgal (2021MAS1009)
Swati Tomar (2021MAS1025)

Supervised by:- Dr. Arvind Kumar

FUZZY LOGIC CONTROL

CONTENT

1. Basic Introduction Of FLC
2. Components of FLC
3. Application of FLC
4. Rule Base
5. Defuzzification
6. Conclusion
7. References

OBJECTIVE

Our objective in this presentation is to build a model on Automatic Door Using fuzzy logic for saving heat energy.

Basic Introduction Of FLC

- Fuzzy logic control (FLC) is well known artificial intelligence, based control technique .
- It utilizes the prior experience of the functionary about the system to be controlled.
- The main role of the functionary is to set up decision-based rules by analyzing the system behavior and the linguistic input variables within the framework of the system.
- The inputs provided to the FLC have to process through three basic stages
  1. Fuzzification,
  2. Decision-making stage,
  3. Defuzzification

  before generating the output

Components of FLC

Fuzzifier

This block transforms input information to fuzzy logic data format that can express with linguistic. The blurred variables obtained are called as linguistic.

Knowledge Base

Normalization of parameters involved partitioning of input and output spaces , selection of membership function of primary fuzzy sets. Knowledge base consists of data base and rule base.
IF (condition (one or more)**) THEN (action)**

Design of FLC

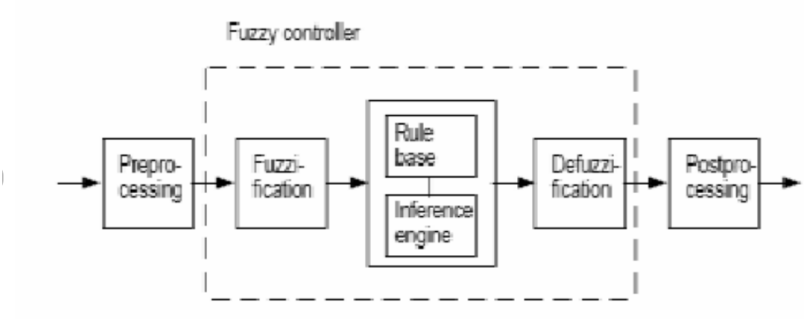


Figure: The structure of fuzzy logic controller

Continued

Inference Engine

This engine generates blurred results by applying the inputs coming from fuzzifier on the knowledge base rules.

In this study we use mamdani inference method.

Defuzzifier

The role of defuzzifier is to convert fuzzy values into crisp values getting from fuzzy inference engine.

Steps for designing Of FLC

- Identification of variable
- Fuzzy subset configuration :- The universe of information spanned by each variable is divided into number of fuzzy subsets and each subset is assigned a linguistic variable.
- Obtaining membership function
- Fuzzy rule base configuration :- Formulate fuzzy rule base by assigning relationship between fuzzy input and output.

- Normalizing and scaling factors:- Appropriate scaling factors for input and output variables must be chosen to normalize variables between $[0, 1]$ and $[-1, 1]$.
- Fuzzification
- Identification of Output
- Defuzzification

Methods to Control Automatic Door

- Field Programmable Gate Array (FPGA)
- Peripheral Interface Controller (PIC) microcontroller.
- Fuzzy Control method(FLC)
- Programmable Logic Controller (PLC) and many more.

Application of FLC

Fuzzy Logic Based Automatic Door

- Fuzzy logic based an automatic door control system is designed to provide for heat energy savings.
- The heat energy loss usually occurs in where automatic doors are used.
- Designed fuzzy logic system's Input statuses (WS: Walking Speed and DD: Distance Door) and the Output status (DOS: Door Opening Speed) is determined.

Identification of Variables

Input Variables :-

(WS: Person's Walking Speed [0-50 m/min]).

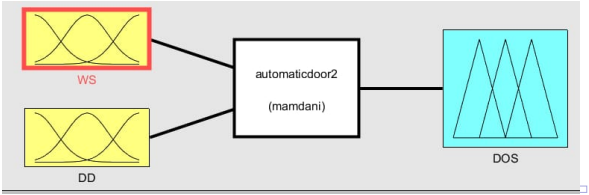
WS= VS (Very Slow) – S (Slow) – M (Medium) – F (Fast) – VF (Very Fast)

(DD: The Distance to Door of person [0-6 m]).

DD= VC (Very Close) – C (Close) – M (Medium) – A (Away) – TA (Too Away)

Output Variables :- (DOS: The Door Opening Speed [0-500 rpm]).

WS= VS (Very Slow) – S (Slow) – M (Medium) – F(Fast) – VF(Very Fast)



Example

Input variables		Output variables
WS(m/min)	**DD (m)**	**DOS (rpm)**
VS [0-12,5]	VC [0-1,5]	VS [0-150]
S [0-25]	C [0-3]	S [0-250]
M [12,5-37,5]	M [1,5-4,5]	M [125-375]
F [25-50]	A [3-6]	F [250-500]
VF [37,5-50]	TA [4,5-6]	VF [375-500]

4/9

Table1.Input parameters, output parameters, degree of the membership function and linguistic expressions of designed fuzzy controller

Rule Base

The fuzzy Controller decides according to the rules contained in the rule base that created with help of an expert. In this study 25 rules are created by experts using membership functions.

	DD					
WS	*TA*	*A*	*M*	*C*	*VC*	
VS	*VS*	*VS*	*S*	*S*	*M*	
S	*VS*	*S*	*S*	*M*	*F*	
M	*S*	*S*	*M*	*F*	*F*	***DOS***
F	*S*	*M*	*F*	*F*	*VF*	
VF	*M*	*F*	*F*	*VF*	*VF*	

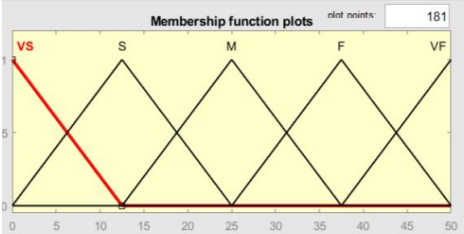


Figure: Input WS

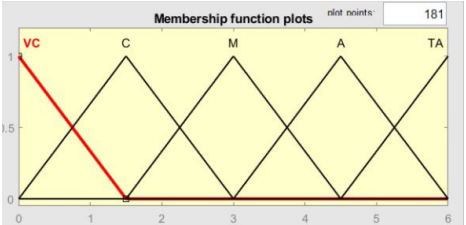


Figure: Input DD

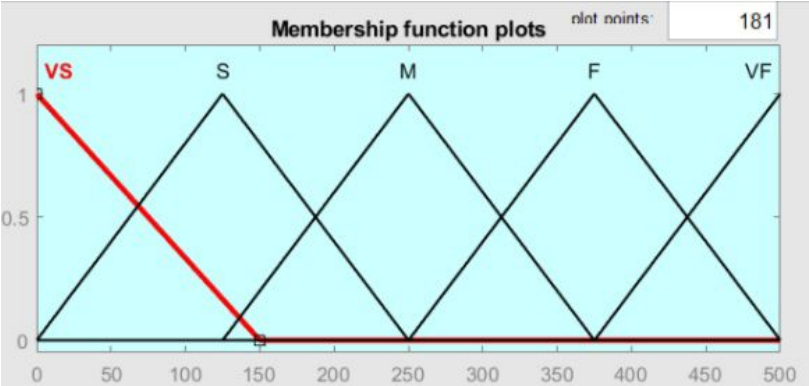


Figure: Output DOS

Defuzzification:-

In this study, we used the center-of-gravity/area (centroid) defuzzification method. In this method, the output of each membership functions and the corresponding maximum membership value z^* are calculated by the formula given below.

$$z^* = \frac{\sum \mu(\bar{z}).\bar{z}}{\sum \mu(\bar{z})}$$

where $\bar{z}$ is the distance to the centroid of the respective membership functions.

Surface

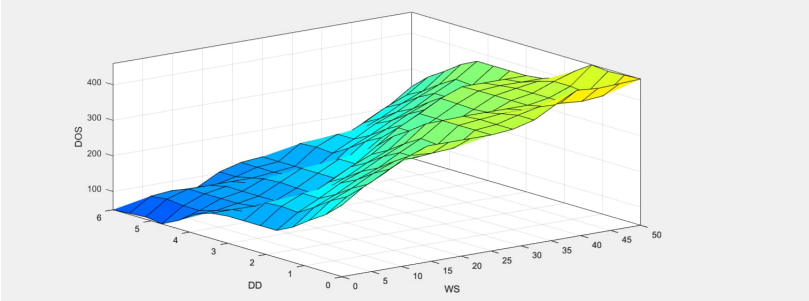


Fig.Input/output response surface viewer

Example

Command Window

New to MATLAB? See resources for Getting Started.

```
>> test = readfis('automaticdoor2.fis');
>> evalfis(test,[29.6,3.19])

ans =

  277.8050

>>
```

Figure: Ex.1

New to MATLAB? See resources for Getting Started.

```
>> test = readfis('automaticdoor2.fis');
>> evalfis(test,[12.5,2])

ans =

  204.5504

>>
```

Figure: Ex.2

Example

Command Window

New to MATLAB? See resources for Getting Started.

```
>> test = readfis('automaticdoor2.fis');
>> evalfis(test,[10,1.5])

ans =

  219.8276

>>
```

Figure: Ex.3

Working Rule

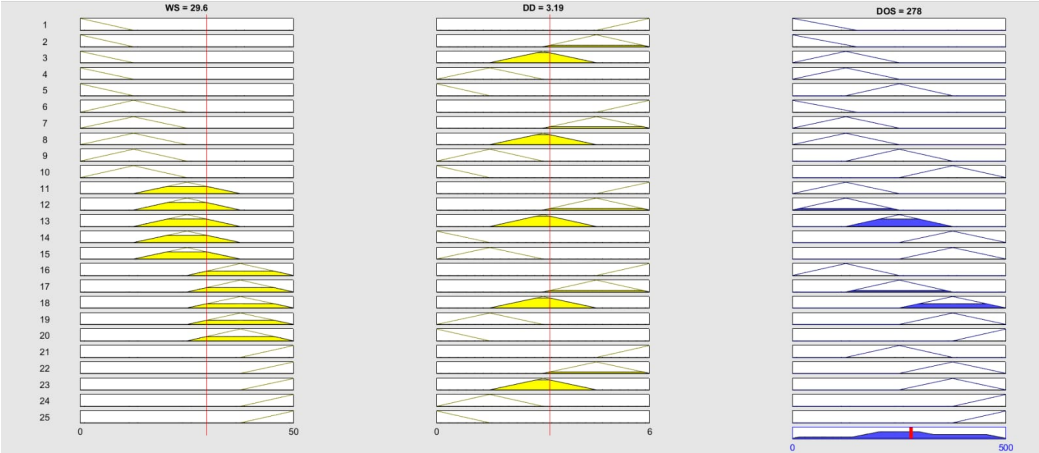


Figure: Detection of DOS

Continued

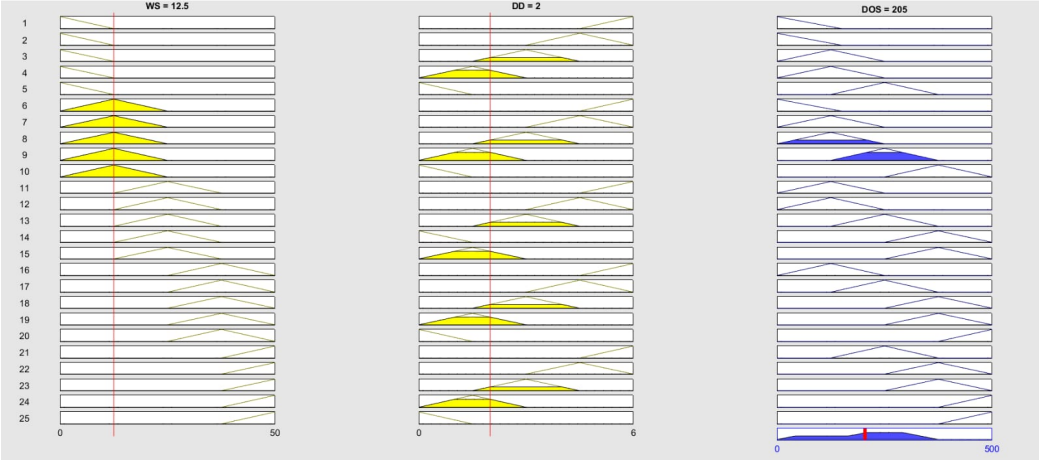


Figure: Detection of DOS

Conclusion

With Fuzzy

- The door speed is adjusted to open as required range.
- People do not wait in front of the door when they come.
- There is less hot air to out from the inside (because of the door don't opened until the end and the door range decreased). Thus saving energy would be provided,

Continued

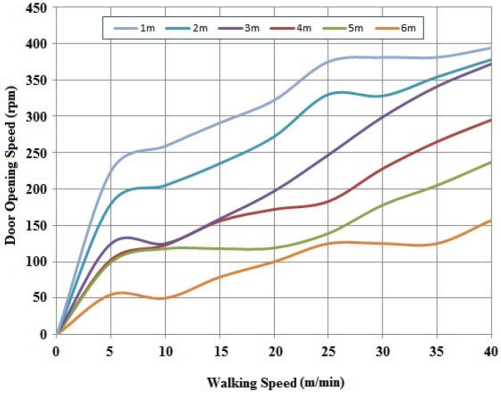


Figure: Variation of door opening speed according to the person's walking speed and distance to door

Continued

Without Fuzzy

- In classical systems, an electronic sensor (Passive Infrared sensor (PIR) sensor) is placed in front of an automatic door and the door opens or closes according to the information from PIR and the person's distance to the door and the walking speed of the person have no effect on the opening speed of the door.
- The door do not open until the point in where the person perceives by the sensor and the person may have to wait in front of the door for a while. At that point, the door opens at constant speed .

References

- Harun SUMBUL; FUZZY LOGIC-BASED AUTOMATIC DOOR CONTROL SYSTEM, *Carpathian Journal of Electrical Engineering*, Vol. 11, **(2017)**.
- Youtube-1 : `https://youtu.be/l7LnN_gzWLI`.
- Youtube-2 : `https://youtu.be/aer5eYXARlg`.

Thank You